Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The energy stored in four capacitors, each of capacitance ' C', placed in parallel configuration and connected to a ' V' volt source is

(A) CV

(B) $\mathrm{CV}^2/2$

(C) $(1/8)\mathrm{CV}^2$

(D) $2\mathrm{CV}^2$

Q.2

The separation between a parallel plate capacitor having capacitance $2\ \mathrm{F}$ is $0.5\ \mathrm{cm}$. The area of the plates of capacitor is

(A) $3.03 \times 10^9\ \mathrm{m}^2$

(B) $1.13 \times 10^9\ \mathrm{m}^2$

(C) $3.13 \times 10^9\ \mathrm{m}^2$

(D) $0.13 \times 10^9\ \mathrm{m}^2$

Q.3

Two capacitors of capacitances $0.3\mu\mathrm{F}$ and $0.6\mu\mathrm{F}$ are connected in series across 6 volts. The ratio of energies stored in them will be

(A) $2:1$

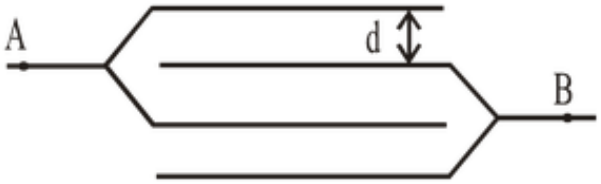
(B) $1:2$

(C) $2:3$

(D) $4:1$

Q.4

The capacitance of arrangement of 4 plates of area A at distance d as shown in figure is



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(A) $\frac{A\varepsilon_0}{d}$

(B) $\frac{3\ A\varepsilon_0}{d}$

(C) $\frac{2\ A\varepsilon_0}{d}$

(D) $\frac{4\ A\varepsilon_0}{d}$

Q.5

Three capacitors of $2\mu F, 3\mu$ F and 6μ F are connected in series to a $10\ V$ source. The charge on the middle capacitor is

(A) $10\mu F$

(B) $5\mu F$

(C) $1\mu F$

(D) Zero

Q.6

A parallel plate capacitor with plate area $100\ cm^2$ and separation between the plates $1.0\ cm$ is connected across a battery of emf $24\ V$. The force of attraction between the plates is

(A) $1.0 \times 10^{-7}\ N$

(B) $2.5 \times 10^{-7}\ N$

(C) $4 \times 10^{-5}\ N$

(D) $1.6 \times 10^{-5}\ N$

Q.7

A capacitor having capacitance $50\mu F$ and initial charge $5mC$ is connected across a resistor at $t = 0$. After one second the potential difference between the plates of the capacitor becomes $70\ V$. Then at $t = 2sec$.

(A) 24% of initial electrostatic energy will remain in the capacitor.

(B) 20% of initial electrostatic energy will remain in the capacitor.

(C) 49% of initial electrostatic energy will remain in the capacitor.

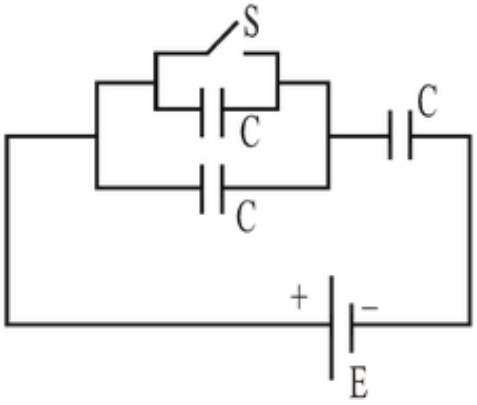
(D) 60% of initial electrostatic energy will remain in the capacitor.

Q.8

Three capacitors each having capacitance $C = 1\mu F$ are connected as shown in figure with a battery of emf

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6 V. If the switch is closed



(A) the amount of the charge flowing through the cell will be $2\mu C$

(B) the heat generated in the circuit is $6\mu J$

(C) the amount of the charge flowing through the cell will be $3\mu J$

(D) the energy absorbed by the battery after closing the switch is $3\mu J$

Q.9

A parallel plate capacitor is charged to a potential V using a battery which is later disconnected.
A dielectric slab is inserted between the plates such that it completely fills the space between the plates. If A is the area of the plates and d is the plate separation, then work done in inserting the dielectric is (K is the dielectric constant)

(A) $\frac{\varepsilon_0 A}{2Kd} v^2$

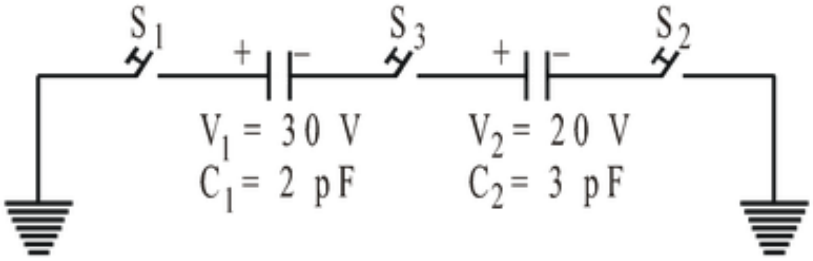
(B) $\frac{\varepsilon_0 A}{2 K^2 d} v^2$

(C) $\frac{\varepsilon_0 A}{2 d}\left(1-\frac{1}{K}\right) v^2$

(D) $\frac{\varepsilon_0 A}{2 d}\left(1-\frac{1}{K^2}\right) v^2$

Q.10

For the circuit shown in figure. Which of the following statements is true?



(A) With S_1 closed, $V_1 = 15$ V, $V_2 = 20$ V

(B) With S_1 closed, $V_1 = V_2 = 25$ V

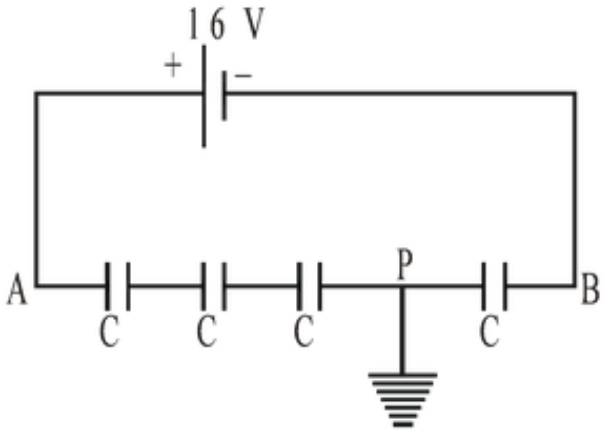
(C) With S_1 and S_2 closed $V_1 = V_2 = 0$

(D) with S_1 and S_3 closed $V_1 = 30$ V and $V_2 = 20$ V

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Q.11

Four identical capacitors are connected in series with a battery of $16\ \mathrm{V}$ between A and B as shown in figure. If the point P is earthed, then the potential at A and B is



(A) $16\ \mathrm{V}, 0\ \mathrm{V}$

(B) $12\ \mathrm{V}, -12\ \mathrm{V}$

(C) $12\ \mathrm{V}, -4\ \mathrm{V}$

(D) $8\ \mathrm{V}, -8\ \mathrm{V}$

Q.12

A parallel plate capacitor is made by stacking n equally spaced plates connected alternatively. If the capacitance between any two adjacent plates is C, then the resultant capacitance is

(A) c

(B) $(n+1)c$

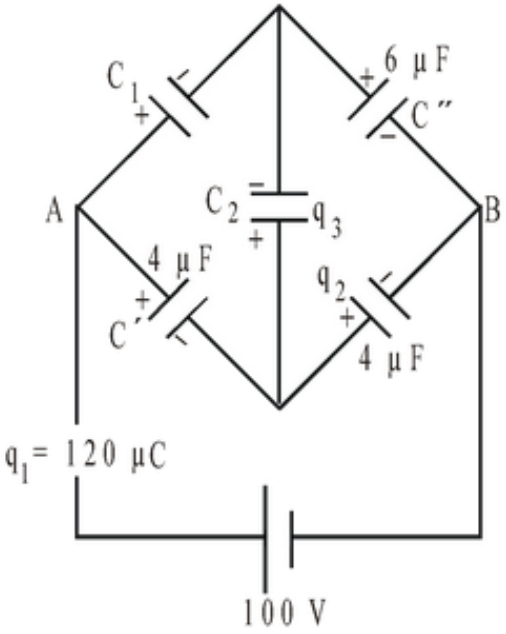
(C) nc

(D) $(n-1)c$

Q.13

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In the circuit shown



(A) $|q_2| = 480\mu C$

(B) $|q_3| = 160\mu C$

(C) $|q_2| = 120\mu C, q_3 =$ zero

(D) It is impossible to find q_2 and q_3 unless C_1 and C_2 are known

Q.14

A capacitor when filled with a dielectric $K = 3$ has charge Q_0, voltage V_0 and field E_0. If the dielectric is replaced with another one having $K = 9$ after disconnecting it with the battery the new values of charge, voltage and field will be respectively

(A) $3Q_0, 3\ V_0, 3E_0$

(B) $Q_0, 3\ V_0, 3E_0$

(C) $Q_0, \frac{V_0}{3}, 3E_0$

(D) $Q_0, \frac{V_0}{3}, \frac{E_0}{3}$

Q.15

A parallel plate capacitor having a plate separation of $2\ mm$ is charged by connecting it to a $300\ V$ supply. The energy density is

(A) $0.01\ J/m^3$

(B) $0.1\ J/m^3$

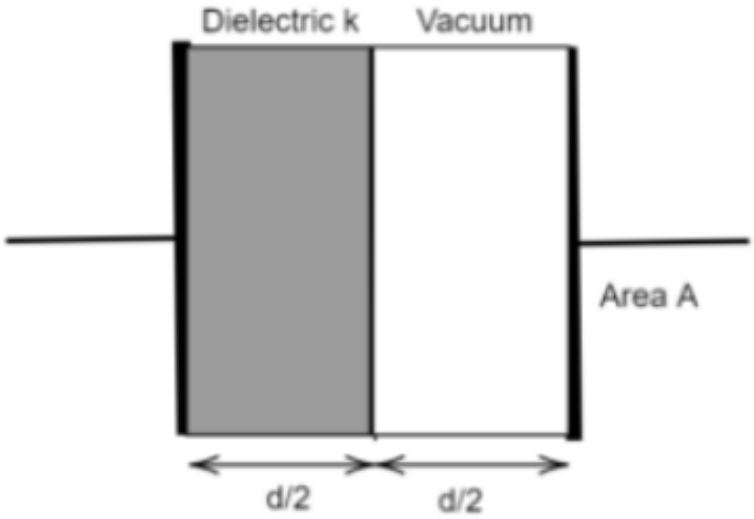
(C) $1.0\ J/m^3$

(D) $10\ J/m^3$

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Q.16

A parallel plate air capacitor has a capacitance C. When it is half filled with a dielectric of dielectric constant 5, the percentage increase in the capacitance will be



(A) 400%

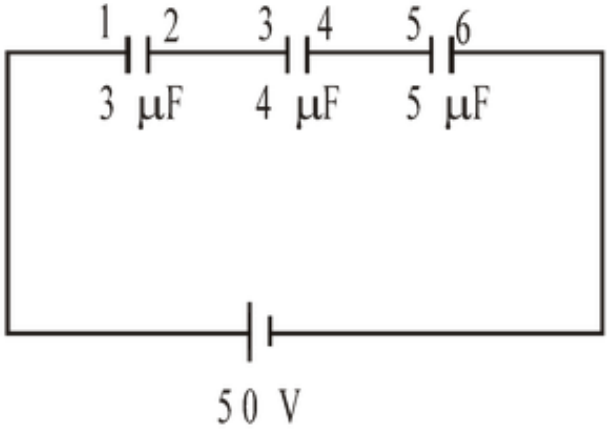
(B) 66.6%

(C) 33.3%

(D) 200%

Q.17

Three capacitors are connected in series to a dc source, as shown in figure. The charges accumulated on the plates of the capacitor are q_1, q_2, q_3, q_4, q_5 and q_6. Then



(A) $q_2 + q_4 + q_6 = \frac{50}{12}\text{C}$

(B) $q_1 + q_3 + q_5 = \frac{50}{12}\text{C}$

(C) $q_2 + q_4 + q_6 = 0$

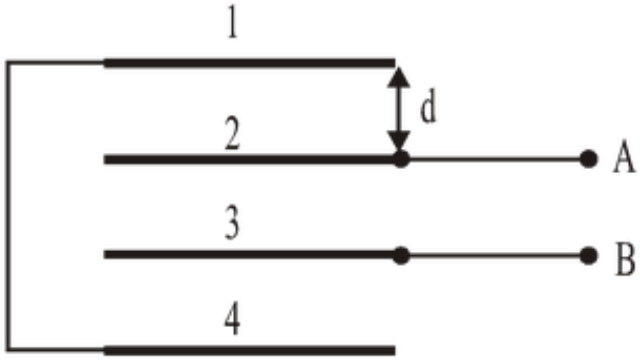
(D) $q_2 = q_4 = q_6$

Q.18

Four metallic plates of equal surface area A are arranged, as shown in figure. The equivalent capacitance

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between A and B is,



(A) $\frac{\varepsilon_0 A}{d}$

(B) $\frac{3\varepsilon_0 A}{d}$

(C) $\frac{3\varepsilon_0 A}{2d}$

(D) $\frac{2}{3}\frac{\varepsilon_0 A}{d}$

Q.19

A capacitor of capacity C is charged to a potential difference V and another capacitor of capacity $2\mathrm{C}$ is charged to a potential difference $4\ \mathrm{V}$. The charging batteries are disconnected and the two capacitors are connected with reverse polarity (i.e. positive plate of first capacitor is connected to negative plate of the second capacitor and the negative plate of first capacitor is connected to positive plate of second capacitor). The heat produced during the redistribution of charge between the capacitor will be

(A) $\frac{125CV^2}{3}$

(B) $\frac{50CV^2}{3}$

(C) $2\mathrm{CV}^2$

(D) $\frac{25CV^2}{3}$

Q.20

A huge capacitor of capacitance is charged to a huge voltage V and then discharged through a dc motor. The motor raises a mass m through h comparable to earth's radius R. The capacitor energy is fully converted to the gravitational potential energy. If $h = R/2$, then V is equal to

(A) $\sqrt{\frac{mgR}{2C}}$

(B) $\sqrt{\frac{mgR}{C}}$

(C) $\sqrt{\frac{2mgR}{C}}$

(D) None of these

Q.21

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In a parallel plate capacitor, the capacitance increases from $6\mu\mathrm{F}$ to $60\mu\mathrm{F}$ on introducing a dielectric medium between the plates. The dielectric constant of the medium is

Q.22

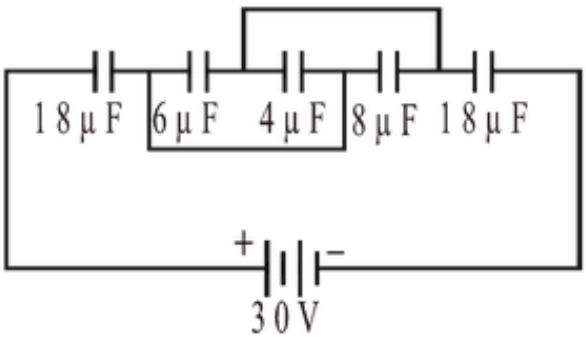
An air capacitor of capacity $\mathrm{C} = 10\mu\mathrm{F}$ is connected to a constant voltage battery of $12\ \mathrm{V}$. Now, the space between the plates is filled with a liquid of dielectric constant $(n) = 5$. The charge that now flows from battery to the capacitor in micro coulomb is

Q.23

Let C be the capacitance of a capacitor discharging through a resistor R. Suppose t_1 is the time taken for the energy stored in the capacitor to reduce to half its initial value and t_2 is the time taken for the charge to reduce to one-fourth its initial value. Then the ratio of $\frac{t_1}{t_2}$ will be

Q.24

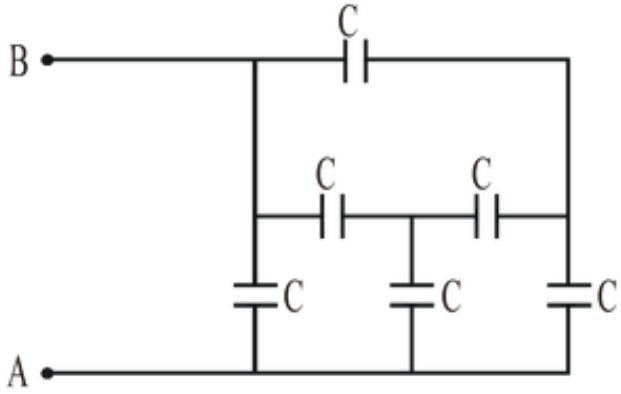
Five capacitors are connected to each other as shown below. What is the potential drop and charge across $4\mu\mathrm{F}$ capacitor



What is charge in Mirco coloumb ?

Q.25

Six capacitors, each of capacitance C, are connected as shown in figure. The equivalent capacitance between A and B is,



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Q.26

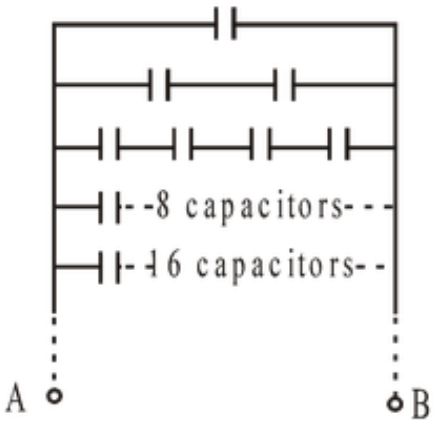
A parallel-plate air capacitor is connected to a battery. While connected to the battery, a dielectric medium whose dielectric constant is 2 is inserted, filling the space between the plates. The energy stored in the capacitor before the dielectric is inserted is E_0. The energy stored is the capacitor after the dielectric has been inserted is E_k. The ratio E_k/E_0 is

Q.27

A $40\mu\text{F}$ capacitor in a defibrillator is charged to $3000\ \text{V}$. The energy stored in the capacitor is sent through the patient during a pulse of duration $2\ \text{ms}$. The power delivered to the patient is ____ (in kW)

Q.28

An infinity number of identical capacitors each of capacitance $1\mu\text{F}$ are connected as shown in the figure. Then the equivalent capacitance between A and B is : Mark answer in μC



Q.29

A photographic flash unit consists of a xenon filled tube. It gives a flash of average power $2000\ \text{W}$ for $0.04\ \text{s}$. The flash is due to discharge of a fully charged capacitor of 40 μF. The voltage to which it is charged before a flash is given by the unit is

Q.30

A parallel plate capacitor with air as the medium between the plates has a capacitance of $10\mu\text{F}$. The area of the capacitor is divided into two equal halves and filled with two media having dielectric constant $\text{k1} = 2$ and $\text{k2} = 4$ the capacitance will now be?

Answers & Solutions

Q.1 Answer:

2CV^2

Solution:

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As the capacitance of all the capacitors is equal(C) so, total capacitance in parallel, $\mathrm{C}' = \mathrm{nC}$

(n = number of capacitors in parallel)

So, $C' = 4C$

Total energy stored $= \frac{1}{2}\mathrm{C}'\mathrm{V}^2$

$= \frac{1}{2}\left(4\mathrm{CV}^2\right)$

$= 2\mathrm{CV}^2$

Q.2 Answer:

$1.13 \times 10^9\ \mathrm{m}^2$

Solution:

$\mathrm{C} = 2\ \mathrm{F}; \mathrm{d} = 0.5 \times 10^{-2}\ \mathrm{m}; \varepsilon_0 = 8.85 \times 10^{-12}\ \mathrm{F/m}$

$\mathrm{C} = \frac{\varepsilon_0\ \mathrm{A}}{\mathrm{d}}$

$\mathrm{Or}, \quad \mathrm{A} = \frac{\mathrm{Cd}}{\varepsilon_0}$

$= \frac{2 \times 0.5 \times 10^{-2}}{8.85 \times 10^{-12}}\ \mathrm{m}^2$

$= 1.13 \times 10^9\ \mathrm{m}^2$

Q.3 Answer:

$2 : 1$

Solution:

$\mathrm{C1} = 0.3\mu\mathrm{f}$

$\mathrm{C2} = 0.6\mu\mathrm{f}$

$\frac{\mathrm{C}_1}{\mathrm{C}_2} = \frac{0.3\mu\mathrm{F}}{0.6\mu\mathrm{F}}$

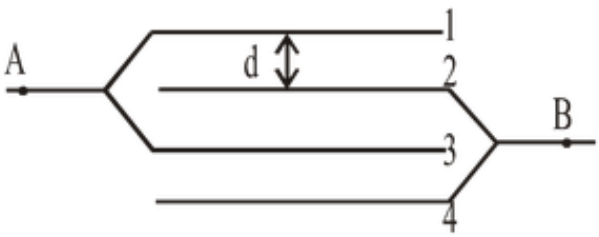
$= \frac{1}{2}$

In series combination of capacitors $\mathrm{U} = \frac{\mathrm{Q}^2}{2\mathrm{C}}$ **So,** $\frac{\mathrm{U}_1}{\mathrm{U}_2} = \frac{\mathrm{Q}^2/2\mathrm{C}_1}{\mathrm{Q}^2/2\mathrm{C}_2} = \frac{\mathrm{C}_2}{\mathrm{C}_1} = \frac{2}{1}$

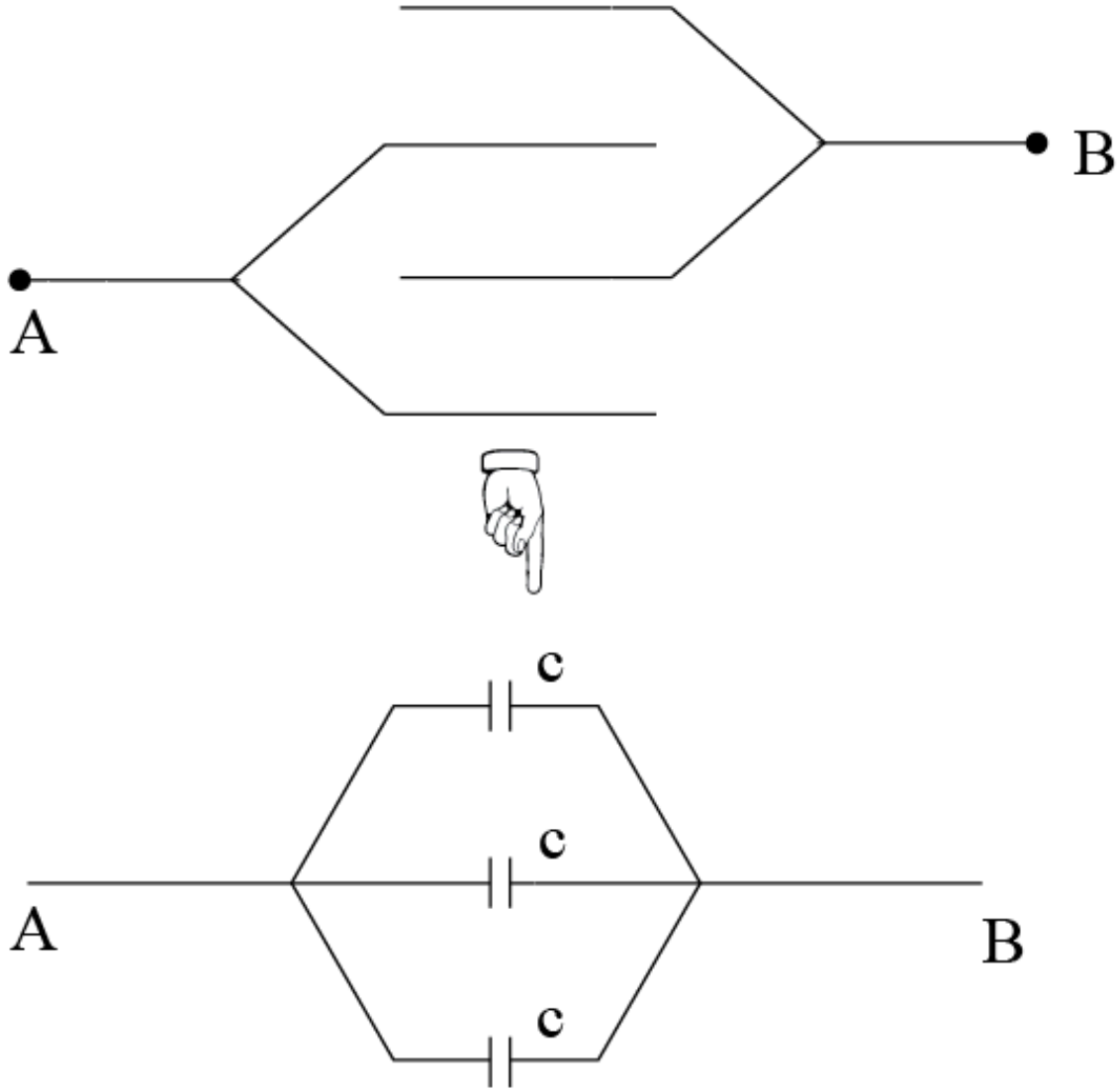
Q.4 Answer:

$\frac{3\ \mathrm{A}\varepsilon_0}{\mathrm{d}}$

Solution:



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$$c' = c + c + c = 3c$$
$$= \frac{3 \times \varepsilon_0 A}{d}$$

Plate 1 and 2 make one set of capacitors. Similarly plate 2 and 3 make one set of capacitor and plate 3 and 4 make one set of capacitor. Hence there are three capacitors joined in parallel.

Hence equivalent capacitance is:

$C = \frac{3\ A\varepsilon_0}{d}$

Q.5 Answer:

$10\mu F$

Solution:

For series combination,

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$
$$\frac{1}{C} = \frac{1}{2\mu F} + \frac{1}{3\mu F} + \frac{1}{6\mu F}$$

or, $C = 1\mu F$

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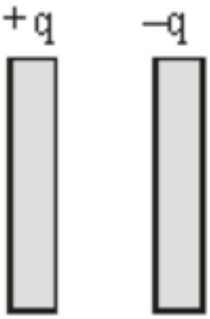
$Q = CV$
$= 1\mu F \times 10$
$= 10\mu C$

Since, charge on each capacitor is same in series combination, the charge on the $3\mu F$ capacitor is also $10\mu C$.

Q.6 Answer:

2.5×10^{-7} N

Solution:



Force between the two plates of capacitor $C = A\varepsilon/d$

$F = \frac{Q^2}{2\varepsilon_0 A} = \frac{\left(\frac{\varepsilon_0 A}{d} \cdot V\right)^2}{2\varepsilon_0 A}$
$= \frac{\varepsilon_0^2 A^2 V^2}{d^2 2\varepsilon_0 A} = \frac{\varepsilon_0 A V^2}{2 d^2}$
$= \frac{8.85\times10^{-12}\times100\times10^{-4}\times24\times24}{2\times10^{-4}} = 2.5 \times 10^{-7}$ N

Q.7 Answer:

24% of initial electrostatic energy will remain in the capacitor.

Solution:

Initial Voltage $= Q/C = 5mc/50$ **microfarad** $= 100$ V
$70 = 100e^{-2/t}$
$V = 100e^{-1/t}$
i.e. $V = 70e^{-1/2}$
Hence $V = \frac{70\times70}{100} = 49\%$
$U\% = \frac{\frac{1}{2}CV^2}{\frac{1}{2}CV_0^2} \times 100 = \frac{49^2}{100^2} \times 100 = 24\%$

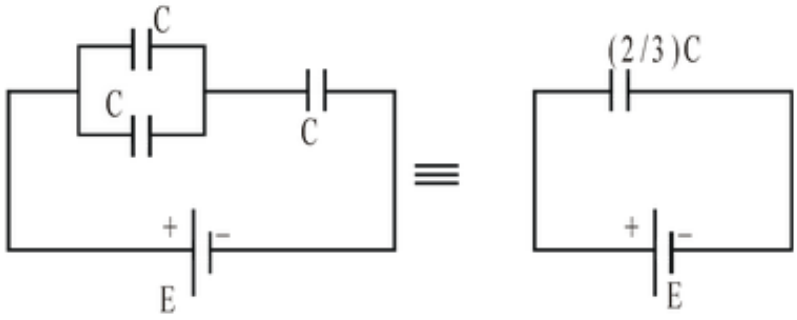
Q.8 Answer:

the heat generated in the circuit is $6\mu J$

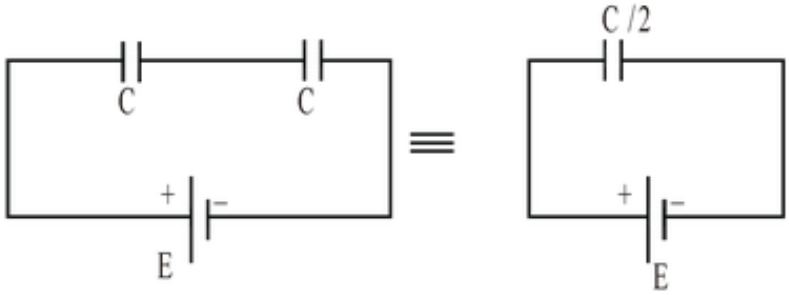
Solution:

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When switch is open



When switch is closed



Total charge $q_1 = \frac{2}{3}CE$

$q_2 = \frac{1}{2}CE$

Hence charge flowing through the cell

$\Delta q = \frac{2}{3}CE - \frac{1}{2}CE = \frac{1}{6}CE$

$\Delta q = \frac{1}{6}cv = 1\mu C$, **Hence work done on the battery** $W = \Delta qE = 6\mu J$ **and heat generated in the circuit** $= 6\mu J$

Q.9 Answer:

$\frac{\varepsilon_0 A}{2d}\left(1 - \frac{1}{K}\right)v^2$

Solution:

Before insert dielectric, capacitance, $C_0 = \frac{A\epsilon_0}{d}$

Magnitude of charge on each plate, $Q = C_0 V = \frac{A\epsilon_0}{d} V$

Before disconnect the battery, filed between the plates, $E_0 = \frac{V}{d}$

Afe insert dielectric, electric filed, $E = \frac{E_0}{K} = \frac{V}{Kd}$

Before insert dielectric, work done, $= W_i = \frac{1}{2}C_0 V^2$

After insert work done, $W_f = \frac{1}{2}(KC_0)\left(\frac{V}{K}\right)^2 . \left(C_f = \frac{A\epsilon_0 K}{d},\ V_f = \frac{E}{d}\right)$

So net work done, $W = W_i - W_f = \frac{1}{2}C_0 V^2 - \frac{1}{2}(KC_0)\left(\frac{V}{K}\right)^2 = \frac{1}{2}C_0 V^2\left(1 - \frac{1}{K}\right) = \frac{A\epsilon_0 V^2}{d}\left(1 - \frac{1}{K}\right)$

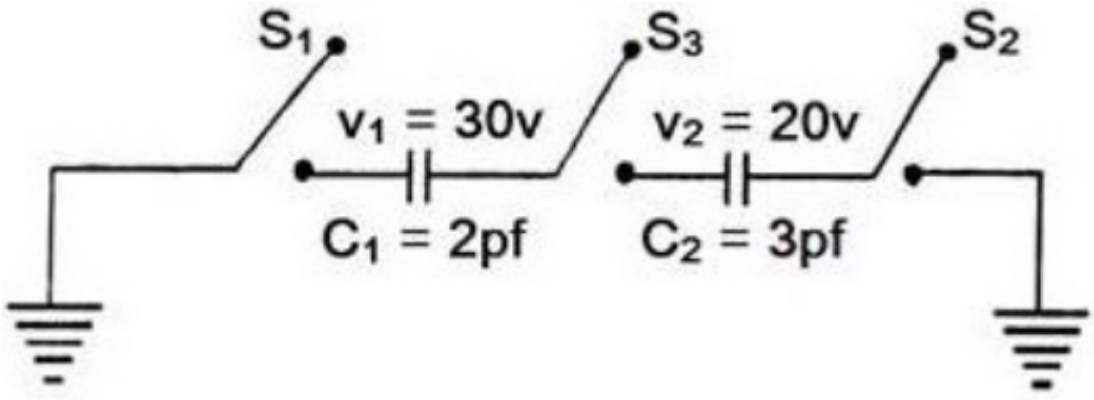
Q.10 Answer:

with S_1 and S_3 closed $V_1 = 30$ V and $V_2 = 20$ V

Solution:

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When switch S_3 is closed, the charge on c1 and c2 will same so no charge drop occur so v1 and v2 will remain same



Q.11 Answer:

$12\ \text{V}, -4\ \text{V}$

Solution:

The p.d. across each capacitor is $4\ \text{V}$. Since there are 3 capacitors between the point A and the ground, the potential difference between A and ground is $12 + 0 = 12\ \text{V}$, the potential difference between ground and the point B is $0 - 4 = -4\ \text{V}$.

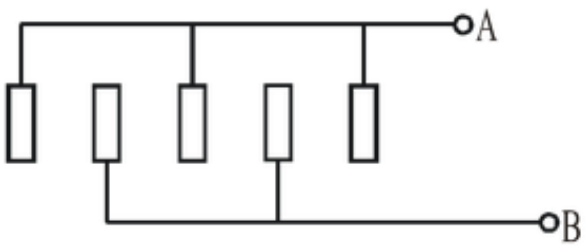
The potential at A is $12\ \text{V}$ and that at B is $-4\ \text{V}$

Q.12 Answer:

$(n-1)c$

Solution:

Each plate will take part in the formation of two capacitors except the plate at the ends. All capacitors will be in parallel and n plates will form $(n-1)$ capacitors.



Q.13 Answer:

$|\text{q}_3| = 160\mu\text{C}$

Solution:

$\because$ potential difference across

$\text{C}' = \frac{120\mu\text{C}}{4\mu\text{F}} = 30\ \text{V}$

Therefore potential drop across C'' must be $70\ \text{V}$ because net potential difference between A and B is $100\ \text{V}$. Then charge on C''

$\text{q}_2 = 4 \times 10^{-6} \times 70 = 280\mu\text{C}$

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The three plates of C_2, C' and C'' which are connected with one another forms an isolated system therefore net charge on this system must be zero.

Therefore,

$$-120\mu\mathrm{C} + 280\mu\mathrm{C} + \mathrm{q}_3 = 0$$
$$\therefore \mathrm{q}_3 = 160\mu\mathrm{C}$$

Q.14 Answer:

$Q_0, \frac{V_0}{3}, \frac{E_0}{3}$

Solution:

When there is no battery, charge remains same while potential difference and electric field decreases i.e. $Q' = Q_0, V' = \frac{V_0 \times 3}{9} = \frac{V_0}{3}$ **and**
$E' = \frac{E_0 \times 3}{9} = \frac{E_0}{3}$

Q.15 Answer:

$0.1\ \mathrm{J/m^3}$

Solution:

The energy density of parallel plate capacitor given by $U = \frac{1}{2}\varepsilon_0 E^2 = \frac{1}{2}\varepsilon_0 \left(\frac{V}{d}\right)^2$
$= \frac{1}{2} \times 8.85 \times 10^{-12} C^2/Nm^2 \times \left(\frac{300 volt}{2\times10^{-3} m}\right)$
$= 0.1\ \mathrm{J/m^3}$

Q.16 Answer:

66.6%

Solution:

Initial capacitance $= \frac{\epsilon \mathrm{A}}{\mathrm{d}}$

When it is half filled by a dielectric of dielectric constant k **then**

$$\mathrm{C}_1 = \frac{\mathrm{ke,A}}{\frac{\mathrm{d}}{2}} = 2\mathrm{k}\frac{\epsilon \mathrm{A}}{\mathrm{d}}$$
$$\text{and } \mathrm{C}_2 = \frac{\epsilon,\mathrm{A}}{\frac{\mathrm{d}}{2}} = \frac{2\varepsilon,\mathrm{A}}{\mathrm{d}}$$
$$\frac{1}{\mathrm{C}} = \frac{1}{\mathrm{C}_1} + \frac{1}{\mathrm{C}_2}$$
$$= \frac{\mathrm{d}}{2\mathrm{e}\mathrm{A}}\left(\frac{1}{\mathrm{k}} + 1\right)$$
$$= \frac{\mathrm{d}}{2\epsilon,\mathrm{A}}\left(\frac{1}{5} + 1\right)$$
$$= \frac{\mathrm{d}}{2\epsilon,\mathrm{A}}\left(\frac{6}{5}\right)$$
$$= \frac{3\,\mathrm{d}}{5\epsilon,\mathrm{A}}$$
$$\therefore \mathrm{C} = \frac{5\epsilon \mathrm{A}}{3\,\mathrm{d}}$$

Hence, % increase in capacitance is

$$= \left(\frac{\frac{\left(\frac{5}{3}\right)\epsilon,\mathrm{A}}{\mathrm{d}} - \frac{\epsilon,\mathrm{A}}{\mathrm{d}}}{\frac{\epsilon \mathrm{A}}{\mathrm{d}}}\right) \times 100 = \frac{2}{3} \times 100 = 66.6\%$$

Q.17 Answer:

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$q_2 = q_4 = q_6$

Solution:

When the capacitors are connected in series, same amount of charge (q) gets deposited on the plates of the capacitor.

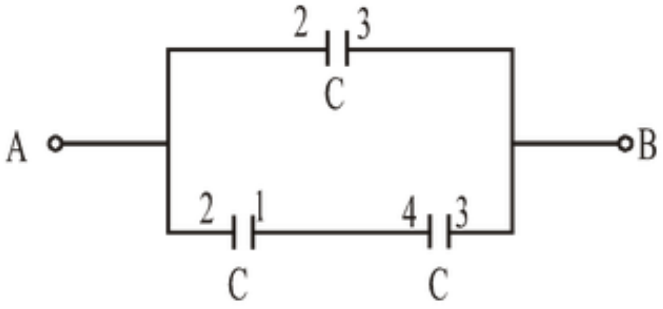
The charge distribution on the plates is as shown in figure. From the figure it can be seen that the charge developed $q_2 q_4 q_6$ on the plates 2,4 and 6 is the same.

Q.18 Answer:

$\frac{3\varepsilon_0 A}{2d}$

Solution:

The equivalent circuit for the given combination of plates is shown in figure. The effective capacitance of the series combination of C_1 and C_2 is



$$\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{1}{\frac{\varepsilon_0 A}{d}} + \frac{1}{\frac{\varepsilon_0 A}{d}}$$

$$C_s = \frac{\varepsilon_0 A}{2d}$$

C_s and C_3 are parallel.

$C_{AB} = C_s + C_3 = \frac{\varepsilon_0 A}{2d} + \frac{\varepsilon_0 A}{d}$

$\Rightarrow C_{AB} = \frac{3}{2}\frac{\varepsilon_0 A}{d}$

Q.19 Answer:

$\frac{25CV^2}{3}$

Solution:

Initial energy of the system,

$U_i = \frac{1}{2}CV^2 + \frac{1}{2} \cdot 2C(4V)^2 = \frac{33}{2}CV^2$

Final charge on the two capacitors after connections are $\frac{7CV}{3}$ and $\frac{14CV}{3}$ Final energy of system

$U_f = \frac{1}{2C}\left(\frac{7CV}{3}\right)^2 + \frac{1}{2.2C}\left(\frac{14CV}{3}\right)^2 = \frac{49CV^2}{6}$

$\therefore$ Heat produced $= U_i - U_f = \frac{25}{3}CV^2$

Q.20 Answer:

$\sqrt{\frac{mgR}{2C}}$

Solution:

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The capacitor energy is $\frac{1}{2}\mathrm{CV}^2$ and GPE is $\left(\frac{mgh}{1+\frac{h}{R}}\right)$

So, we have $\frac{mgh}{1+\frac{h}{R}} = \frac{1}{2}CV^2$

If $h = R/2$, then $mgR/4 = \frac{1}{2}CV^2$. This implies that $v = \sqrt{\frac{mgR}{2C}}$

Q.21 Answer:

—

Solution:

After introducing the dielectric medium, the capacitance of the capacitor is $C' = \mathrm{kC}$ (Where C is the capacitance of the air-filled capacitor)

$$60\mu\mathrm{F} = \mathrm{k}(6\mu\mathrm{F})$$
$$\Rightarrow \mathrm{k} = \frac{60\mu\mathrm{F}}{6\mu\mathrm{F}}$$
$$= 10$$

Q.22 Answer:

—

Solution:

Initial charge $= 10\mu\mathrm{F} \times 12\ \mathrm{V} = 120\mu\mathrm{C}$ New capacitance $\mathrm{C}' = \mathrm{nC}$

$\& = 5 \times 10\mu\mathrm{F} = 50\mu\mathrm{F}$

New charge $= \mathrm{VC}'$

$\& = 50\mu\mathrm{F} \times 12\ \mathrm{V} = 600\mu\mathrm{C}$

So, charge flowing now from battery to the capacitor Q' = Final charge - initial charge

$= (600 - 120)\mu\mathrm{C} = 480\mu\mathrm{C}$

Q.23 Answer:

—

Solution:

$\mathrm{U} = \frac{1}{2}\frac{\mathrm{q}^2}{\mathrm{C}} = \frac{1}{2\mathrm{C}}\left(\mathrm{q_0 e^{-t/T}}\right)^2 = \frac{\mathrm{q_0^2}}{2\mathrm{C}}\mathrm{e^{-2t/T}}$

(where $\tau = \mathrm{CR}$)

$\mathrm{U} = \mathrm{U_i e^{-2t/\tau}}$

$\frac{1}{2}\mathrm{U_i} = \mathrm{U_i e^{-2t_1/\tau}}$

$\frac{1}{2} = \mathrm{e^{-2t_1/\tau}} \Rightarrow \mathrm{t_1} = \frac{\mathrm{T}}{2}\ln 2$

Now $q = q_0 e^{-t/T}$

$\frac{1}{4}\mathrm{q_0} = \mathrm{q_0 e^{-t/2\ T}}$

$\mathrm{t_2} = \mathrm{T}\ln 4 = 2\ \mathrm{T}\ln 2$

$\therefore \frac{\mathrm{t_1}}{\mathrm{t_2}} = \frac{1}{4} = 0.25$

Q.24 Answer:

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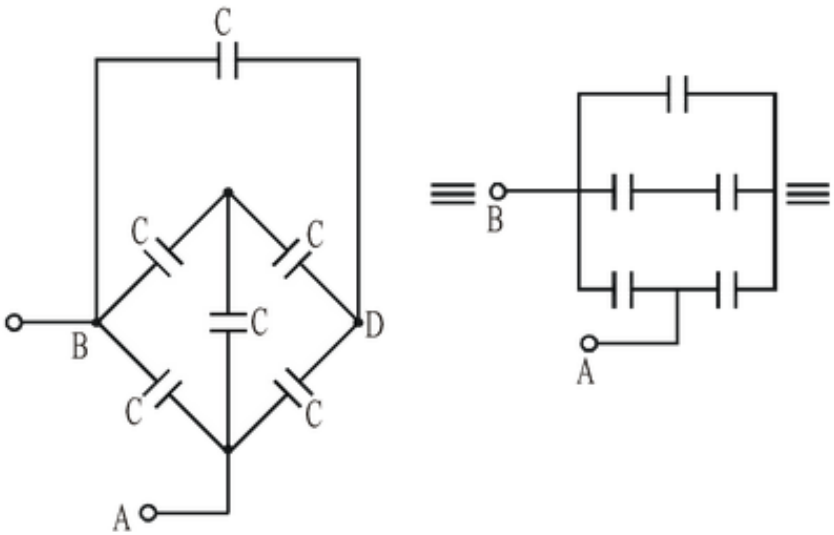
Solution:

The capacitors $6\mu\mathrm{F}, 4\mu\mathrm{F}$ and $8\mu\mathrm{F}$ are in parallel and their capacitance is $18\mu\mathrm{F}$. So, across the power supply, we have three capacitors each of $18\mu\mathrm{F}$ in series.
Hence total capacitance is $\frac{18}{3} = 6\mu\mathrm{F}$.
So the potential drop across each of the $6\mu\mathrm{F}, 4\mu\mathrm{F}$ and $8\mu\mathrm{F}$ capacitors is 10 volts. $\therefore$ charge across $4\mu\mathrm{F}$ capacitor $= 10 \times 4 = 40\mu\mathrm{C}$

Q.25 Answer:

—

Solution:



for whitestone bridge $\mathrm{Ceq} = \mathrm{C}$
again C and C are in parallel combination than total c is $\mathrm{C} + \mathrm{C}$ is 2c

Q.26 Answer:

—

Solution:

Initial energy $\frac{1}{2}\mathrm{CV}^2$
After dielectric medium energy will be $\frac{1}{2}C'\bigvee^2$ and $c' = A\varepsilon/d \quad \varepsilon = \varepsilon E$
u'/u is as :
$E_R = \frac{1}{2}CV^2 = \frac{1}{2}\varepsilon_r C_0 V^2 = \varepsilon_r E_0$
$E_R/E_0 = \varepsilon_r = 2$

Q.27 Answer:

—

Solution:

As we know that work W = Energy stored in it energy as u=
$\frac{1}{2}\mathrm{CV}^2 = 180\ \mathrm{J}$

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Power as $= \mathrm{W/t}$

$$\text{Power } = \frac{\frac{1}{2}CV^2}{t} = \frac{40\times10^{-6}\times(3000)^2}{2\times2\times10^{-3}}$$
$$= 90 \text{ kW}$$

Q.28 Answer:

—

Solution:

This combination form a G.P.

calculation of capacitance as arranged in series combination

$$\frac{1}{\mathrm{C}} = \frac{1}{\mathrm{C_1}} + \frac{1}{\mathrm{C_2}} + \frac{1}{\mathrm{C_3}}$$

$$\text{as } \mathrm{Ca} = \mathrm{C} : \mathrm{Cb} = \mathrm{C}/2; \mathrm{Cc} = \mathrm{C}/4$$

$$\mathrm{S} = 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} \ldots\ldots\ldots$$

Sum of infinite G.P.

$$\mathrm{S} = \frac{a}{1-r}$$

$$\text{Here a} = \text{ first term and r} = \text{ common ratio}$$

$$= \frac{1}{2}$$

$$\frac{1}{1-\frac{1}{2}} = 2 \Rightarrow C_{eq} = 2\mu F$$

Q.29 Answer:

—

Solution:

The total energy stored in the capacitor,

power $\times$ **time** $= 2000 \times 0.04$

Energy can be written as $\frac{1}{2}$ **CV2**

by put values and equal by energy formula

$$2000 \times 0.04 = \frac{1}{2}40 \times 10^{-6}\ \mathrm{V}^2$$

$$\mathrm{V}^2 = 4 \times 10^6 \Rightarrow \mathrm{V} = 2000\ \mathrm{V}$$

Q.30 Answer:

—

Solution:

Initial capacitance is given as $\mathrm{C} = 10\mu\mathrm{F}$

$$\Rightarrow \mathrm{C} = \frac{\varepsilon_0\ \mathrm{A}}{\mathrm{d}} = 10\mu\mathrm{F}$$

New arrangement acts as two capacitors $\mathrm{C_1}$ **and** $\mathrm{C_2}$ **connected in parallel. Thus capacitance of each part** $\mathrm{C_1} = \frac{\mathrm{k_1}\varepsilon_0\ \mathrm{A_1}}{\mathrm{d}}$ **and** $\mathrm{C_2} = \frac{\mathrm{k_2}\varepsilon_0\ \mathrm{A_2}}{\mathrm{d}}$

Equivalent capacitance of parallel combination $\mathrm{C_p} = \mathrm{C_1} + \mathrm{C_2} = (\mathrm{k_1}+$

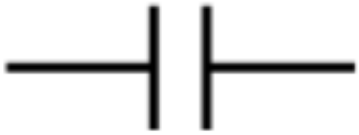
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$k_2) \frac{\epsilon_0(A_1+A_2)}{d}$

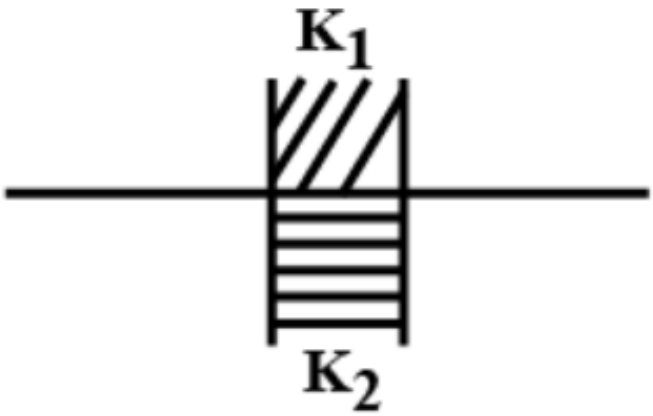
$\Rightarrow C_p = (k_1 + k_2) \frac{\epsilon_0 A}{2d} \quad \left(\because A_1 = A_2 = \frac{A}{2}\right)$

Thus $C_p = (2+4) \times \frac{10}{2} = 30\mu F$

Initial capacitance



Capacitors conneted



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